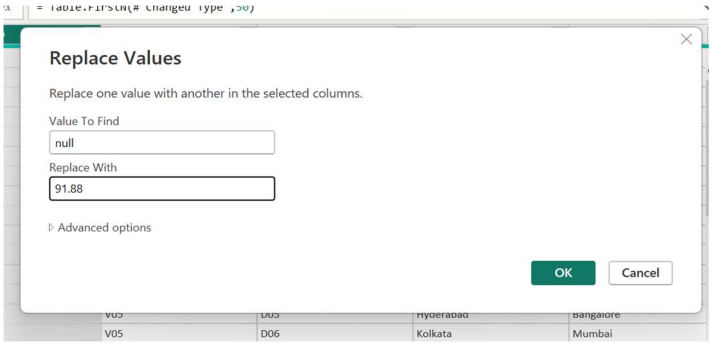
AI-Powered Fleet Performance & Delivery Efficiency Dashboard

1.Data Cleaning

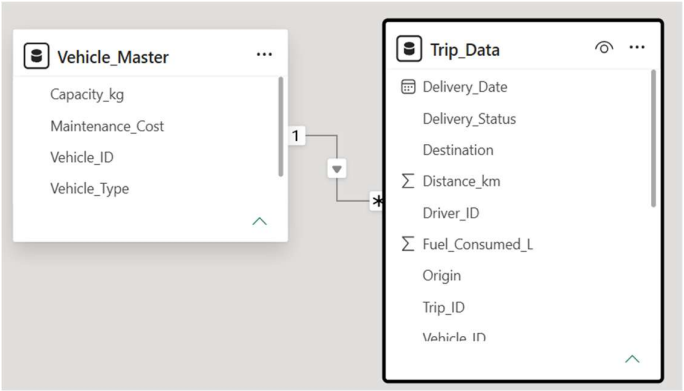
Missing Fuel Consumption Values

Missing values in the **Fuel Consumed (litres)** column of the Trip_Data table were identified and handled using **mean imputation**. The mean of the available fuel-consumption values was calculated using statistic and the missing values were replaced with this mean.



Data Modelling

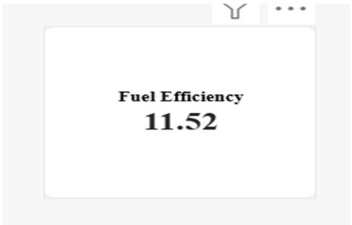
A relationship was created between the **Trip_Data** and **Vehicle Master** tables using the common **Vehicle_ID** field.



2.DAX Measures:

New measure was created to calculate the **Fuel Efficiency** using the DAX,

Fuel Efficiency= DIVIDE(sum(Trip_Data[Distance_km]),SUM(Trip_Data[Fuel_ Consumed _L])). The fuel efficiency was displayed using card visual.



Another measure was created to calculate the on-time delivery (%) using the DAX, **On-Time_Delivery(%) = DIVIDE (CALCULATE(COUNTA(Trip_Data[Delivery_ Status]), Trip_Data[Delivery_ Status]="OnTime"),COUNTA(Trip_Data[Delivery_ Status]))**



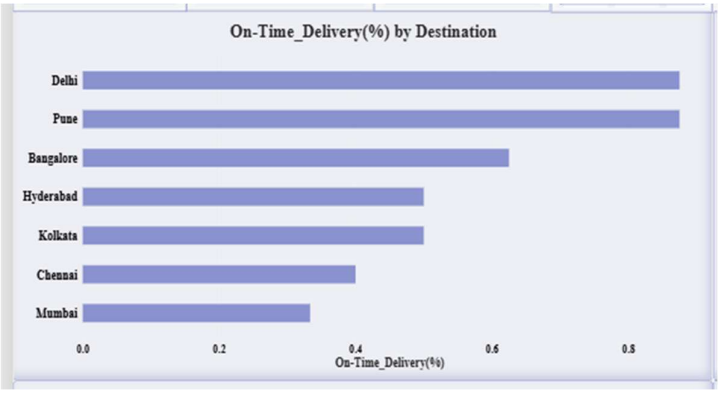
Another measure was created to calculate the cost per km using the DAX**, Cost_per_km =DIVIDE((100+SUM(Vehicle_Master[Maintenance_Cost])),SUM(Trip_ Data[Distance_ km]))**



3. Visualization:

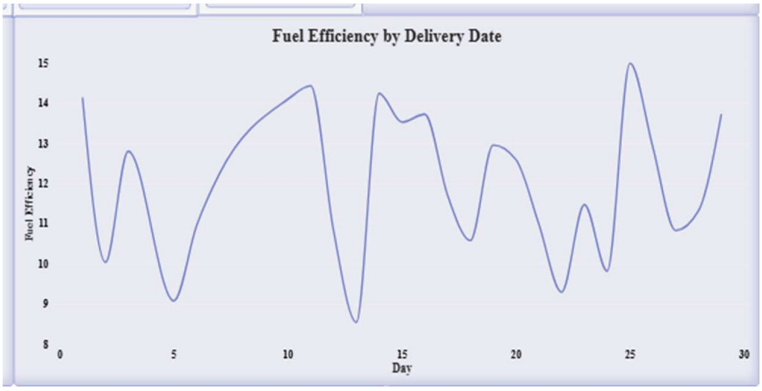
Bar chart:

A clustered bar chart was created using the On-Time Delivery (%) and Destination. It compares the percentage of trips delivered on time for each destination. This helps identify destinations with **strong delivery performance** as well as locations where delays are more frequent.



Line chart:

A line chart was created to compare the Fuel efficiency trend by delivery date. It shows the **trend in fuel efficiency over the delivery dates** in the two-month dataset. This visualization helps identify whether **fuel efficiency is improving or declining over time** and can highlight unusual changes in fleet performance.

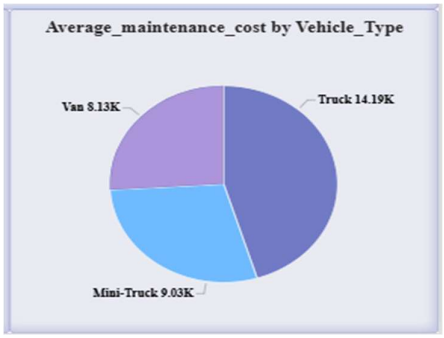


Cards visualization:

Using the card visual Total Trips, On-Time delivery (%), Total distance (km), cost per km and Fuel efficiency was displayed.

Pie chart:

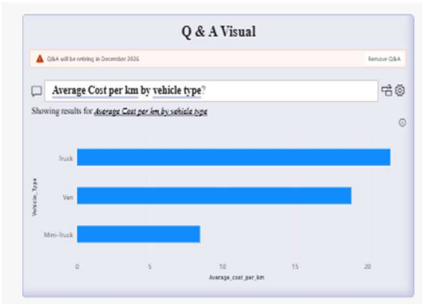
Average maintenance cost was calculated using the DAX, **Average_maintenance_ cost = AVERAGE(Vehicle_Master[Maintenance_Cost]).** A pie chart was created to visualized the vehicle type by average maintenance cost. It helps show how maintenance expenses are distributed among the fleet.



4. AI-Powered Visuals

Q&A Visual:

The Q&A visual was used to analyze the average cost per kilometre for each vehicle type. This helps compare vehicle cost efficiency and identify which vehicle types have higher or lower operating costs per kilometre.



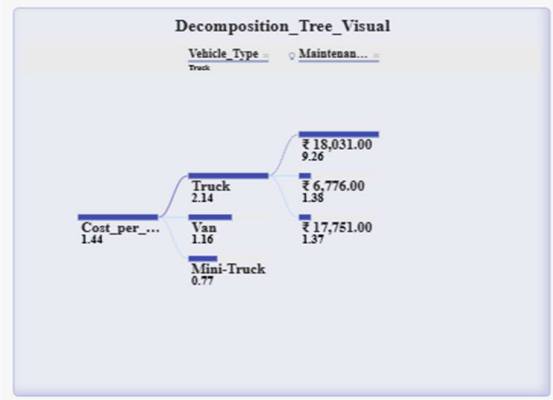
Key Influencers Visual

The Key Influencers visual was used to analyze factors influencing **Delivery Status**, with **Distance (km), Vehicle Type, and Driver ID** as explanatory factors. This analysis clearly shows that delivery status was more likely to be on-time with driver ID DO3 than others on average. It can support the logistics company in identifying factors that contribute to successful or delayed deliveries.



Decomposition Tree

The Decomposition Tree was used to analyze **Cost per km** by breaking it down according to **Vehicle Type, Maintenance Cost, and Distance (km)**. This provides a detailed view of the factors contributing to variations in transportation cost. It helps identify specific vehicle types or combinations of maintenance expenses and travel distances that result in higher costs per kilometre.



For the transport operations dashboard to optimize routes and fleet usage, below charts were created.

- Donut chart for delivery status,
- Bar chart for average capacity by vehicle type
- Donut chart for fuel efficiency by vehicle type
- Bar chart for total distance in km by origin and destination.

Slicer is added to enable users to interactively filter and analyze the dashboard based on Vehicle Type. These visualizations are perfectly arranged in the two dashboards, Transport operations and Fleet & Root Optimization.

